

What is the appropriate data representation of electrochemical impedance spectroscopy in machine-learning analysis?

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ABSTRACT

Electrochemical impedance spectroscopy (EIS) is an important analytic technique for the understanding of electrochemical systems. With the recent advent and burgeoning deployment of machine learning (ML) in EIS analysis, a critical yet hitherto unanswered question emerges: what is the appropriate data representation of EIS for ML-based analysis? While the representation of a model's input data is known to be critical for a successful deployment of ML model, EIS is known to possess multiple classical venues of data representation and it remains unclear how different EIS data should be compared following a proper data normalization protocol. Here we report the methodology and the outcomes that evaluate the efficacy of multiple data representation methods in ML-based EIS analysis. At least within our proof-of-concept parameter space, plotting the input training data's impedance magnitude ($|Z|$) against phase angle (ϕ) while individually normalizing each EIS curve yields the highest accuracy and robustness in the correspondingly established residual neural network (ResNet) model. Rationalized by additional "importance" analysis of the input data, such a data representation method extracts information and hidden features more effectively. While Nyquist plot is more widely used in manual analysis, we found that ML-based analysis may require a different data representation and offered a clear guideline for future researchers to evaluate on a case-by-case basis.

KEYWORDS

Electrochemistry, electrochemical impedance spectroscopy, machine learning, ResNet, data representation, instance normalization, mechanistic analysis

INTRODUCTION

Electrochemical impedance spectroscopy¹ serves as a powerful technique for investigating material properties and analyzing complicated reaction processes in various electrochemical systems, including batteries,²⁻⁴ sensors,⁵⁻⁷ and catalysts.⁸⁻¹⁰ As Figure 1A shows, electrochemical impedance spectroscopy (EIS) data are collected by applying a sinusoidal voltage $E(t)$, consisting of potential perturbation and the bias potential E_S , over a series of frequencies f , and monitoring the corresponding sinusoidal response current $I(t)$ of the system.¹ Impedance Z is defined as the ratio of output signal $E(t)$ and input signal $I(t)$, yielding a complex quantity.^{1, 11, 12} Following validation of impedimetric data, analysis can be conducted through various plotting formats among different components.¹ The most widely used formats are the Nyquist and Bode plots^{1, 11} (Figure 1B), each offering distinct insights into the data: Nyquist plot displays impedance in the complex plane (the absolute value of imaginary part Z'' vs. the real part Z') while Bode plots illustrate the system's response amplitude of impedance $|Z|$ and phase angle shift $|\phi|$ as functions logarithmic value of frequency $\log_{10}(f)$. Diverse visualizations can be employed supplementarily to extract comprehensive insights from data obtained in a single experiment. However, to the best of the author's knowledge, the current trend particularly in mechanistic analysis leans towards the Nyquist plot, attributed to its straightforward distinctive semi-circle visualization.^{1, 13, 14} Taking analyzing aluminum corrosion in concentrated KOH as an example, a series of electrochemical impedance spectroscopy (EIS) data was obtained across various potentials and visualized by Nyquist plot, and the interpretation involves identifying the electrochemical process that best aligns with the loci of experimental impedance spectra, utilizing a set of optimized model parameters.¹⁵ Determining the superior data representation in EIS remains an open question, given the diverse features of various formats, and the challenge of statistically evaluating the impact of data representation on manual EIS interpretation further complicates this inquiry.

The integration of machine learning (ML) in EIS necessitates identifying the optimal data representation for enhanced accuracy and analytical capabilities. ML presents a promising solution for advancing mechanistic analysis in the electrochemical system,¹⁶⁻¹⁹ leveraging its proficiency in uncovering hidden patterns and providing data-driven insights with minimal human interference. Recent advancements showcase ML's capability in automatically analyzing EIS data, specifically for classification of equivalent circuit models and parameter regression.²⁰⁻²² Notably, Chung and

colleagues successfully constructed a deep-learning-based Variational Autoencoders (VAE) model trained on the real and imaginary part of the EIS data,²³ yielding an 82.0% accuracy in classifying five commonly encountered equivalent circuit models, outperforming other ML algorithms including Deep Neural Network,²⁴ Random Forest,²⁰ Support Vector Machine,²² and AdaBoost²⁰ with corresponding accuracy of 78.9%, 58.2%, 48%, and 45.3%, respectively. Despite the extensive exploration of various ML algorithms for automated EIS analysis, there is a current lack of comprehensive evaluation for data preprocessing step, including the aforementioned data representation question and data normalization strategies²⁵. This evaluation is crucial, at least in the context of ML model establishment, as an optimal preprocessing step involving data representation and normalization can significantly enhance the overall performance of ML algorithms.^{26, 27}

To bridge this gap, we seek to answer the question: What is the most appropriate data representation and normalization of electrochemical impedance spectroscopy for machine-learning-aided mechanistic analysis? Here we conducted an apple-to-apple comparison involving two distinct data representations—Nyquist and magnitude-vs-phase plots (derived from Bode plots)—and three normalization strategies—none, layer, and instance normalization (Figure 1C and Figure S1A), employing residual neural networks (ResNet) (Figure 1D) for classifying five common mechanisms in homogeneous molecular electrochemistry as a proof of concept. We evaluate ResNet models' prediction performances under varied data preprocessing conditions (Figure 1E and F), while ensuring consistent machine learning architecture, training processes, hyperparameters, and so on. Additionally, an “importance” evaluation (Figure 1E) illuminates how data preprocessing influences ResNet models' information extraction and decision-making processes, providing crucial insights into fine-tuning data representation and normalization for EIS interpretation.

RESULTS

EIS dataset establishment

The training set for ResNet consists of numerically simulated EIS spectra using the finite-element method (see the Methods section) based on five targeted homogeneous electrochemical mechanisms (Figure 2A): a single-electron transfer with any level of reversibility (E), an E step followed by a C step with any level of reversibility (EC), an E step preceded by a C step (CE), a system of two E steps connected by an irreversible rate-limiting C_i step with the second E step being more thermodynamically facile than the first one (ECE), and a two-electron transfer that is similar to ECE yet the second E step is replaced by a solution disproportionation reaction ($DISP$). The construction of numerical models of PDEs, boundary conditions, and initial conditions follows the mechanisms' definitions in textbooks (see the Supporting Information).^{11, 28} The parameters defining the reaction systems include but not limited to, concentrations, interfacial charge transfer rate constant in the electrochemical process, and thermodynamic equilibrium constants of the chemical process. The constraints and sampling approaches of these parameters follow our group's previous work¹⁸ to ensure physical realism and class diversity. For the parameters incorporated in impedance technique, the number of measured frequencies was set to 500 points, distributed logarithmically in the frequency range (f) from 1 Hz to 100 kHz. Inspired by the reaction mechanism analysis (RMA) based on EIS as a function of applied potentials in the last century,^{15, 29} we collected EIS spectra under different fixed values of applied potentials E_{app} , from $E_0 - 0.4$ to $E_0 + 0.6$ V at 50 or 100 mV intervals for each reaction system, providing more information to potentially distinguish between candidate mechanisms. Approximately 5,000 simulated data points, with fourteen impedance spectra for each data point, were generated for each mechanism by randomly sampling electrochemical parameters within constrained ranges, ensuring a sufficient, balanced and diverse training dataset. Dynamic Gaussian noise was stochastically added on the preprocessed raw data during the training and test processes.

Data representation and preprocessing

The impedance spectra were represented in two ways, Nyquist plot, or magnitude vs phase angle of impedance (Figure 1B), both covering the identical whole information collected by EIS. Similar to the case of cyclic voltammogram,¹⁸ we deployed two-dimensional matrix compatible with ResNet architecture as ML model's input— $\{f, (Z'(E), Z''(E))\}_n$ (n , the number of applied potentials,

f , frequency, Z' and Z'' , the real and imaginary component of impedance respectively) for Nyquist-based representation, and $\{f, \varphi(E), Z(E)\}_n$ (n , the number of applied potentials, f , frequency, φ , phase angle, and Z , the magnitude of impedance) for magnitude-vs-phase-based representation. To determine the optimal normalization methods given the large variations of magnitude in EIS simulated with different electrochemical parameters and applied potentials, different strategies based on min-max scaling were tested for each data format: original data without normalization, layer normalization³⁰ (normalized across the various channels of applied potentials for a single example) or instance normalization³¹ (independently for each applied potential channel for a single example) (see the Supporting Information and Figure S1A). Figure 1C shows the exemplified visualizations of EIS spectra for three different normalization methods per representation. Layer normalization keeps the same relative magnitude between different applied potentials as the original data, while instance normalization introduces greater variations of magnitude across instances or channels. All values are scaled into a specific range between -1 and 1 for both normalization strategies.

Machine learning model construction and evaluation

ResNet (residual neural network), an advanced evolution of convolutional neural networks,³² was chosen for this task, encouraged by previous successful application in extracting features from high-dimensional cyclic voltammograms.¹⁸ ResNet's unique skip connections within the convolutional layers can effectively handle vanishing and exploding gradient problems during the training process, leading to effective learning and fast convergence.³² The two-dimensional input matrix was transformed into a three-dimensional tensor $\{n, y, m\}$ (Figure S1B). In this tensor, n has a dimension of 14 corresponding to the number of simulated applied potentials, m correlates to the number of simulated frequencies (with a dimension of 500). y processes a dimension of 3: the first two channels represent EIS features, $(Z'(E), Z''(E))$ for Nyquist-based representation and $(\varphi(E), Z(E))$ for magnitude-phase-based representation; the third channel contains the values of applied potentials. The yielded output is a vector $y = \{y_1, y_2, y_3, y_4, y_5\}$ (Figure 1E) representing a quantitative distribution of propensity as a surrogate for the statistical probability associated with the respective mechanisms of E , EC , CE , ECE , and $DISP1$. The classification process designates the mechanism with the largest y component as the most probable one for the electrochemical system.

ResNet models, specifically ResNet with 18 residual learning layers, hence ResNet-18, were trained and validated by simulated EIS spectra. We employed two different data representations and three normalization strategies mentioned above during the data preprocessing step, while keeping training process and hyperparameters the same. To account for training randomness, each combination was explored through 1000-epoch training of eight replica ResNet models (Figure S2A). We compared the average accuracy and standard deviation for each combination (Figure 1F and Table S1). Among these six data preprocessing strategies, the magnitude-vs-phase representation with method III normalization yielded the best test accuracies (Figure 2B), with an average accuracy of 89.2% (standard deviation: 1.7%) on the test set. In comparison, the other two magnitude-vs-phase-based models yield less satisfactory results, with an average accuracy of 77.2% (standard deviation: 10.1 %) for method II (Figure 2C) and 70.9% (standard deviation: 11.8 %) for method I (Figure 2D). On the contrary, the ResNet models trained with Nyquist representation yields a 71.8% average accuracy (standard deviation: 1.4%) with method III normalization (Figure 2E), while method II (Figure 2F) and method I (Figure 2G) resulted in lower averages: 68.9% (standard deviation: 6.7%) and 63.7% (standard deviation: 10.2%), respectively. The exemplary confusion matrices of individual models with six preprocessing strategies are illustrated in Figure 2B to 2G.

To better evaluate models' performance, the receiver operating characteristic (ROC) curve was utilized to assess the model's ability to distinguish one electrochemical mechanism from the others, where its corresponding area under the curve (AUC) value quantifies the model's overall discriminatory power. Higher AUC values indicate better discrimination, with a value of 1 indicating perfect performance. In Figure 3A, the optimal model, with magnitude vs phase data representation and method III normalization, shows an excellent classification performance with AUC values above 0.98 for five mechanisms, with an average of 0.99. As a comparison, two suboptimal models show lower AUC values of *DISPI*, *E*, and *EC* mechanisms, corresponding to their confusion matrices (Figure 2C and 2E) with noticeable misclassification these three mechanisms (Figure 3B and 3C). The superior accuracy with magnitude-vs-phase based representation and method III normalization, compared to the others, implies ResNet models' more efficient information extraction from the input, which will be further discussed in subsequent sections.

We also evaluated the robustness of the optimal models against noise, to access its applicability in classifying real-world experimental data. We introduced random dynamic Gaussian noise with varying standard deviation σ (0, 0.01, 0.1) relative to the original data scale into the simulated spectra for the noise-added training set. Subsequently, these models were tested with unseen data containing varying gaussian noise (σ from 0 to 0.3) (Figure S4). The model trained with $\sigma = 0.01$ maintained an overall accuracy of over 87% until $\sigma > 0.1$, and this high accuracy persisted above 85% until $\sigma > 0.25$. Even at training noise $\sigma = 0.1$, the model exhibited an overall accuracy of around 85%, remaining relatively robust up to $\sigma = 0.3$. In contrast, the zero-noise model saw an immediate drop in accuracy, declining from 92.5% to 77% upon the introduction of 0.01 test noise and performed below 60% as noise levels increased beyond 0.1. These findings demonstrate remarkable capacity of the optimal models trained with noise-added data to maintain high accuracy levels even when noise levels exceed the training standard.

As presented previously, introducing training noise enhances robustness with a minor accuracy tradeoff for the magnitude-vs-phase model. To determine the most acceptable noise level for reliable, real-world impedance data, we employed Kramers-Kronig (K-K) transforms,^{33, 34} a mathematical approach to validate the consistency of experimental EIS spectra. Figure S5 shows the exemplary simulated EIS spectra with different noise levels (0, 0.01, 0.1) and their corresponding K-K transform fitting results³⁵. A noisy experimental impedance data can empirically pass the Kramers-Kronig test if the residual errors between the fitted and measured values are similar in size to the measurement noise.¹ As demonstrated in Figure S5B, noise $\sigma = 0.01$ results in residual errors χ^2 at a similar order, with a value of 0.024, suggesting that the expected noise level of validated EIS spectra should be approximately 0.01 of the magnitude. Thus, we argue that the ResNet model trained with $\sigma = 0.01$ is theoretically applicable for experimental data validated by K-K transforms, with the confusion matrix and ROC curves plotted in Figure S6.

DISCUSSION

Encouraged by the enhanced performance when choosing the most appropriate data preprocessing strategies, we aimed to understand the effects of different data preprocessing on model's performance. The "importance" analysis can identify the regions or features in the input that have the most significance influence on the model's decision-making process by visualizing the relative magnitudes of gradient values of the model's output with respect to the input data. For model comparison, the gradient values were collectively normalized based on the highest value achieved among all the ResNet models and plotted as gray color bar in the Nyquist plots. Additionally, for easier visualization of "importance" on each simulated spectrum with varying magnitudes, all simulated spectra were individually normalized, irrespective of the preprocessing methods used for the input data. Taken an *CE*-mechanism system from the test set as a visualization example (Figure 3D), the model based on magnitude-phase representation and method III normalization accurately recognizes an *CE* mechanism with high propensity; the "importance" analysis demonstrates a distribution of "importance" across both low and high frequency regions (Figure 3E), capturing information related to both kinetics and mass transfer, with a particular emphasis on the mass transfer-controlled low-frequency zone. In contrast, on one side, the suboptimal model with magnitude-vs-phase representation and method II normalization misclassified it as *DISP1*, with "importance" plot primarily emphasized on the high frequency region (Figure 3F). Another suboptimal model, the Nyquist-based and method-II-normalized one, finally predicts the system as *ECE* with a relatively higher confidence, whose "importance" exhibits a higher focus on the high frequency region (Figure 4G) predominantly associated with kinetics information. The colored regions in Figure 4H and I illustrate the specific different areas where the optimal model puts more emphasize on than the suboptimal one: red sections indicate higher "importance" in the optimal model, while blue regions signify higher "importance" in the other model. These differences in the models' attention towards various parts of the EIS pattern suggest that the optimal model extracts information and hidden features more effectively, providing an explanation for the significantly improved performance observed. More exemplary results of the "importance" analysis of two models can be found in the supporting information.

Utilizing supporting vector classifier (SVC) models with magnitude-phase and Nyquist-based data representations and the same method III normalization further confirms the enhanced accuracy of

the magnitude-phase approach. The results showed consistent, albeit less satisfactory, performances compared to ResNet models. The magnitude-phase SVC model reached an overall test accuracy of 82.4%, while the Nyquist SVC model scored 66.8%. The consistent improvement in accuracy with the magnitude-phase representation indicates an inherent advantage of such representation, particularly suited for the specific mechanistic analysis task conducted in this study, instead of model bias.

CONCLUSIONS

In this work, we answered what the most appropriate data representation and normalization of electrochemical impedance spectroscopy is for machine-learning-aided mechanistic analysis, using ResNet model and a classification task of five mechanisms as a proof-of-concept. The optimal ResNet ensemble model with phase-magnitude-based data representation, method III normalization achieved an 89.2% average test accuracy and an average area under the curve of the receiver operating characteristic (AUC-ROC) value of 0.99. Trained with 0.01 dynamic Gaussian noise, the ResNet model demonstrates robustness beyond the noise tolerance validated by Kramers-Kronig transformation, yielding an overall test accuracy exceeding 87.3%. Our study underscores the necessity of exploring multiple data representation and preprocessing strategies, at least in the context of ResNet-aided EIS-based mechanistic analysis, suggesting the superior data representation even in human inspection. This study also lays the foundation for improved model accuracy, setting a precedent for enhanced mechanistic analysis and classification tasks in diverse electrochemical systems, such as heterogeneous and catalytic processes. We envision that this approach will continually pave the ways for automated EIS-based mechanistic analysis, making enhanced EIS interpretation more accessible to a broader research community.

METHODS

Finite-element simulation of electrochemical impedance spectra

Finite-element simulations of electrochemical impedance spectroscopy (EIS) spectra were conducted using COMSOL Multiphysics v5.6. The Electrochemistry Module and electroanalysis physics were employed to establish a one-dimensional model under the inert supporting electrolyte assumption. A time-dependent solver specialized for alternating current (AC) impedance was incorporated, using an adaptive mesh with a maximal mesh size of one twenty-fifth of the diffusion layer thickness at the maximum frequency. For impedance spectrum simulation, a harmonic perturbation was applied to an assigned applied potential E_{app} over a range of frequencies. Simulations were iterated using COMSOL Multiphysics 5.6 with MATLAB R2022b, with model parameters sampled by Python 3 scripts and communicated to COMSOL through MATLAB. Additional sanitization was implemented after COMSOL simulation to ensure electrochemical feasibility. We conducted approximately 30,000 valid simulations, each with up to fourteen different applied potential values (E_{app}) in the EIS spectra. Supporting Information contains detailed model information and variable constraints for each mechanism type.

Establishment of machine learning algorithm

Two data representation strategies and three normalization approaches were employed for data preprocessing, as reported in the main text (Figure 2C) before the implementation of machine learning. For each data point comprised of simulated EIS spectra at n number of E_{app} values, EIS spectra were preprocessed and translated into the two-dimensional matrix $\{f, (Z'(E), Z''(E))\}_n$ or $\{f, (\varphi(E), Z(E))\}_n$ for two different representations, Nyquist and magnitude-vs-phase-angle plot, respectively. For layer normalization, we initially normalized the simulated values ($Z'(E), Z''(E)$) or $(\varphi(E), Z(E))$ by subtracting the minimum value across different applied potentials and dividing it by the range (maximum – minimum values) among those potentials. Then, the values were scaled from the range of 0 to 1 to the range of –1 to 1 by multiplying the normalized value by 2 and subtracting 1. In the instance normalization, a similar procedure was applied, but the values were independently normalized using the maximum and minimum values at each applied potential. Original values, without any normalization strategies, were utilized as input to the ML algorithms in the non-normalization case for comparative analysis. Dynamic Gaussian noise was

stochastically added on the preprocessed raw data during the training and test processes.

Machine-learning code was implemented on PyCharm IDE using Python3 code. PyTorch machine-learning frameworks were used to implement residual neural networks (ResNet). Scikit-learn frameworks were used to implement supporting vector classifier (SVC). Graphs were generated using the Matplotlib library and the Pyplot module. Each data point was processed using python API OpenCV into a three-dimensional tensor $\{n, y, m\}$ with a size of $\{14 \times 3 \times 500\}$ as an input. Here n has a dimension of 14 correlating to the number of simulated applied potentials E_{app} , and m has a dimension of 500 correlating to the number of simulated frequencies f . y has a dimension of 3, where two channels for the EIS features, $(Z'(E), Z''(E))$ for Nyquist-based representation and $(\phi(E), Z(E))$ for magnitude-vs-phase-angle based representation, and the last channel for the value of applied potentials E_{app} . EIS features were sequenced from the lowest to highest frequency, though the information of the frequency values is not provided to the machine learning model. The yielded output is a vector $y = \{y_1, y_2, y_3, y_4, y_5\}$, a quantitative distribution of propensity as a surrogate of statistical probability toward each of the considered mechanisms of E , EC , CE , ECE , and $DISPI$, respectively. The input tensor was evaluated using a kernel/filter of size $\{14 \times 3 \times 7\}$. The kernel only views the data present in the tensor, and the filter is not changed with different values of n . In terms of the hyperparameters of the ResNet-18 model, the standard learning rate of 1×10^{-3} and standard weight decay of 1×10^{-5} was used for all training. Training was performed on a machine using an Intel i7-12700H 14-Core CPU with 64 GB of RAM and an Nvidia GeForce RTX 3060 GPU. It takes approximately 24 h to train the ResNet model with 1000 epochs and just seconds to predict the class of one sample. Each individual ResNet model underwent 1,000 training epochs to ensure enhanced accuracy, with the training dataset consisting of 3,500 shuffled data points for each mechanism type. The test dataset comprised about 1,500 randomly sampled data points for each mechanism category. For establish the ensemble models with instance normalization, we employed a majority voting approach with eight individual ResNet-18 models by aggregating their results. To evaluate the ensemble model's performance, we utilized a test dataset consisting of 1,000 previously unseen data points for each mechanism. Each simulated EIS spectra were evaluated by the eight individually trained ResNet-18 models, yielding a set of prediction values (y vectors). These y vectors were subsequently averaged and the final prediction were determined based on the maximal component in the y vector.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at,

Protocols of the numerical simulations of electrochemical impedance spectroscopy spectra; detailed definitions of the reaction mechanisms; protocols of establishing the ML models; supplementary tables and figures related to the ML training and evaluation.

AUTHOR INFORMATION

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Author Contributions

C.L. and Q.G. supervised the project. J.S., B.B.H., C.C., and C.L. developed theoretical computational frameworks for the generation of simulated electrochemical impedance spectroscopy spectra. J.S. generated and sanitized the simulated data for model training/test. J.S. developed and implemented various data preprocessing strategies. W.Z. provided the scripts for establishing ResNet models for mechanistic analysis based on simulated cyclic voltammograms. J.S. adapted the algorithms with valuable help from W.Z. and built the ResNet models in this work. J.S. prepared the initial draft of the manuscript. All of the authors discussed the results of the project and assisted with manuscript preparation.

Notes

The authors declare no competing financial interest.

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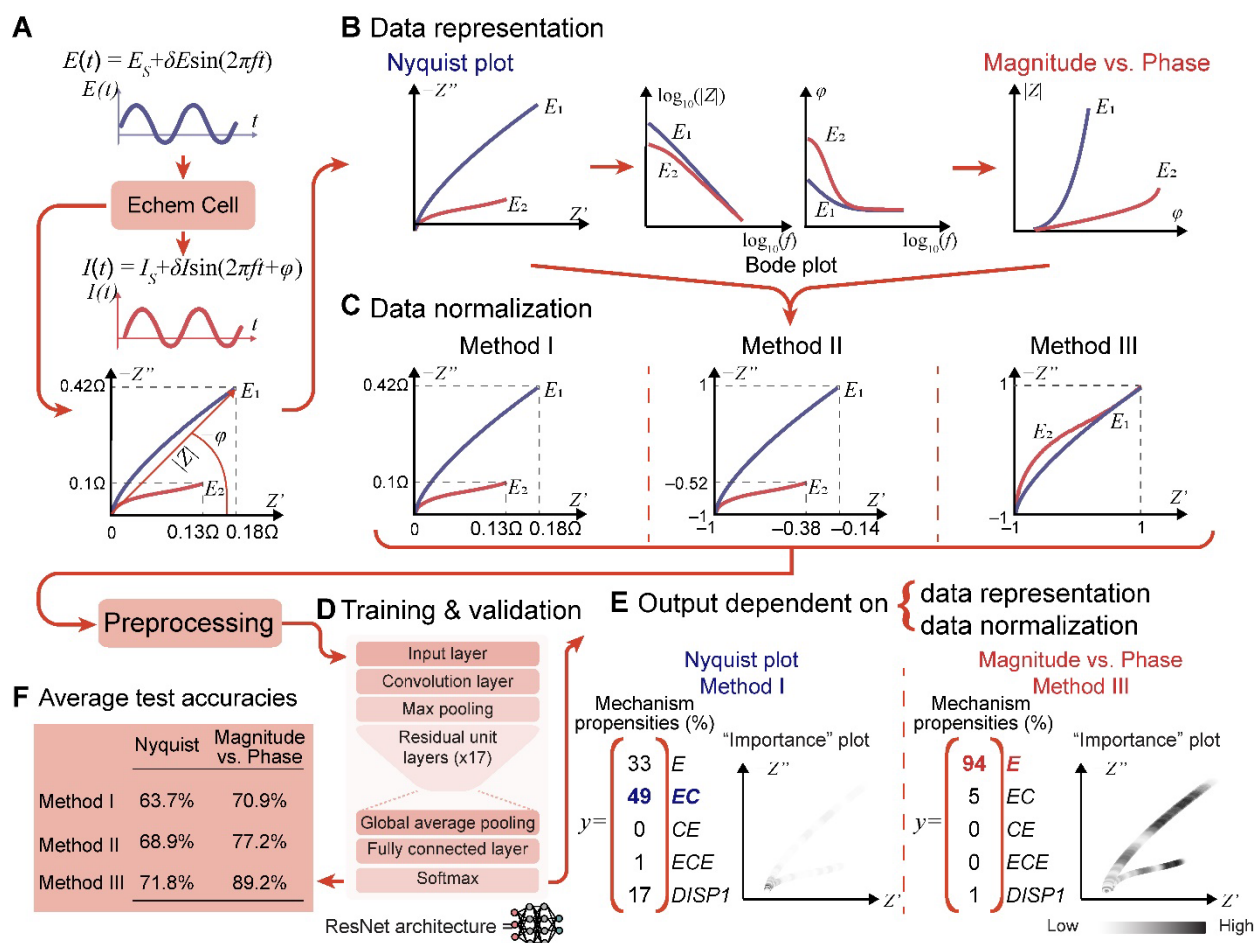


Figure 1. Schematic workflow of this project. (A) Electrochemical impedance spectroscopy (EIS) test, where the electrochemical system is perturbed by a sinusoidal voltage $E(t)$ (consisting of potential perturbation and the bias potential E_s) over a series of frequencies f , yielding alternating current signal $I(t)$. The impedance Z is depicted in the complex plane, where the x axis is the real part Z' and y axis is the imaginary part Z'' . $|Z|$ and φ , magnitude and phase angle of the impedance, respectively. E_1 and E_2 , different bias potentials. (B) Visualization of Nyquist, Bode and magnitude-vs-phase-angle plots for impedance data representation. (C) Three different data normalization methods. (D) Architecture of the residual neural network (ResNet). (E) Output illustration and 'importance' evaluation of models trained with distinct data representation and normalization strategies. (F) Summary of average test accuracies for models trained with six data preprocessing strategies.

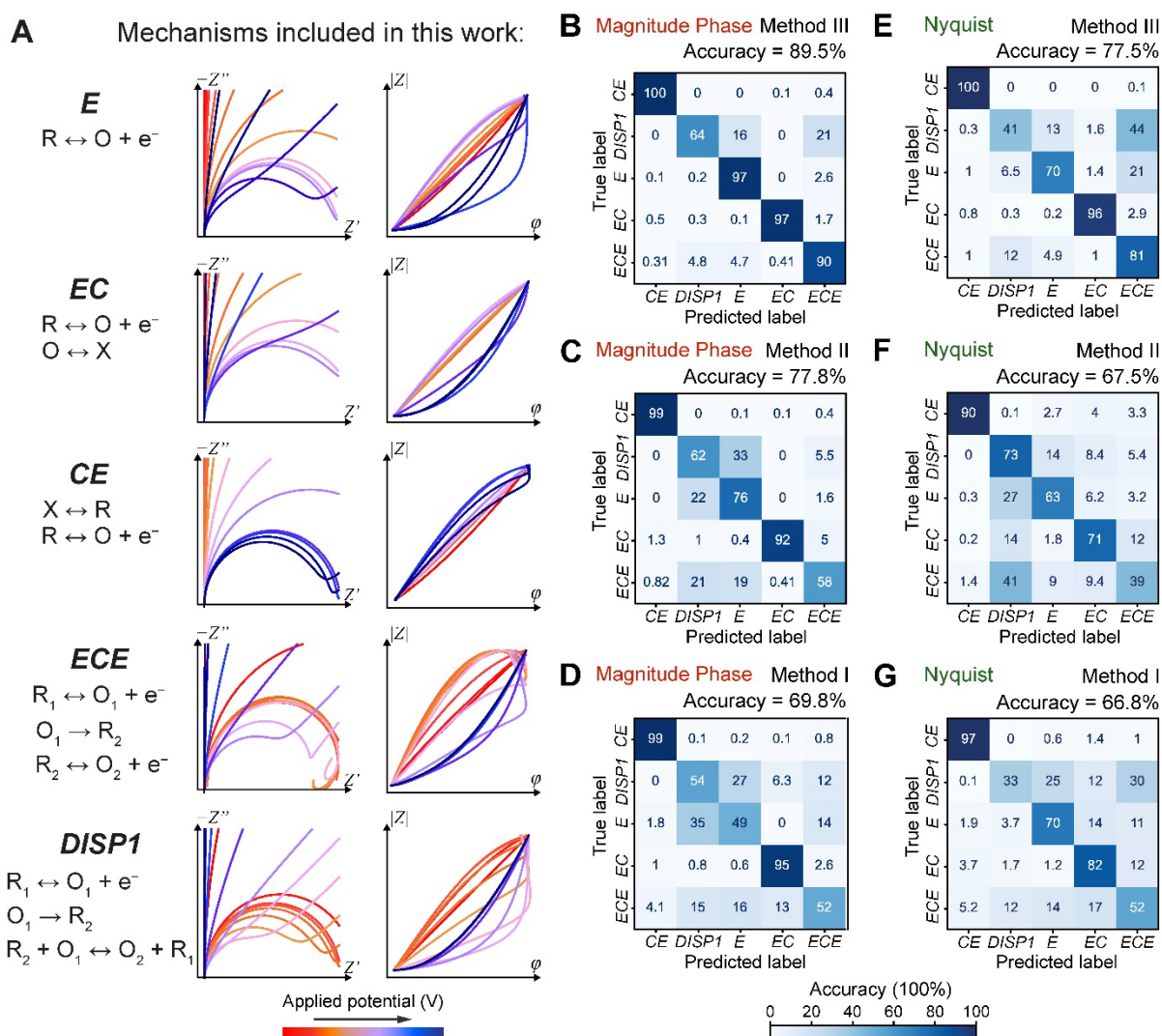


Figure 2. Established ML models for mechanistic analysis with electrochemical impedance spectroscopy (EIS). (A) Five homogeneous molecular electrochemical mechanisms included in this work, normalized EIS spectra visualized in Nyquist (upper row) and magnitude-vs-phase-angle plots (lower row), color-coded for various bias potentials. (B) to (G) Exemplary confusion matrices depicting individual ResNet models' performance with six distinct data preprocessing strategies.

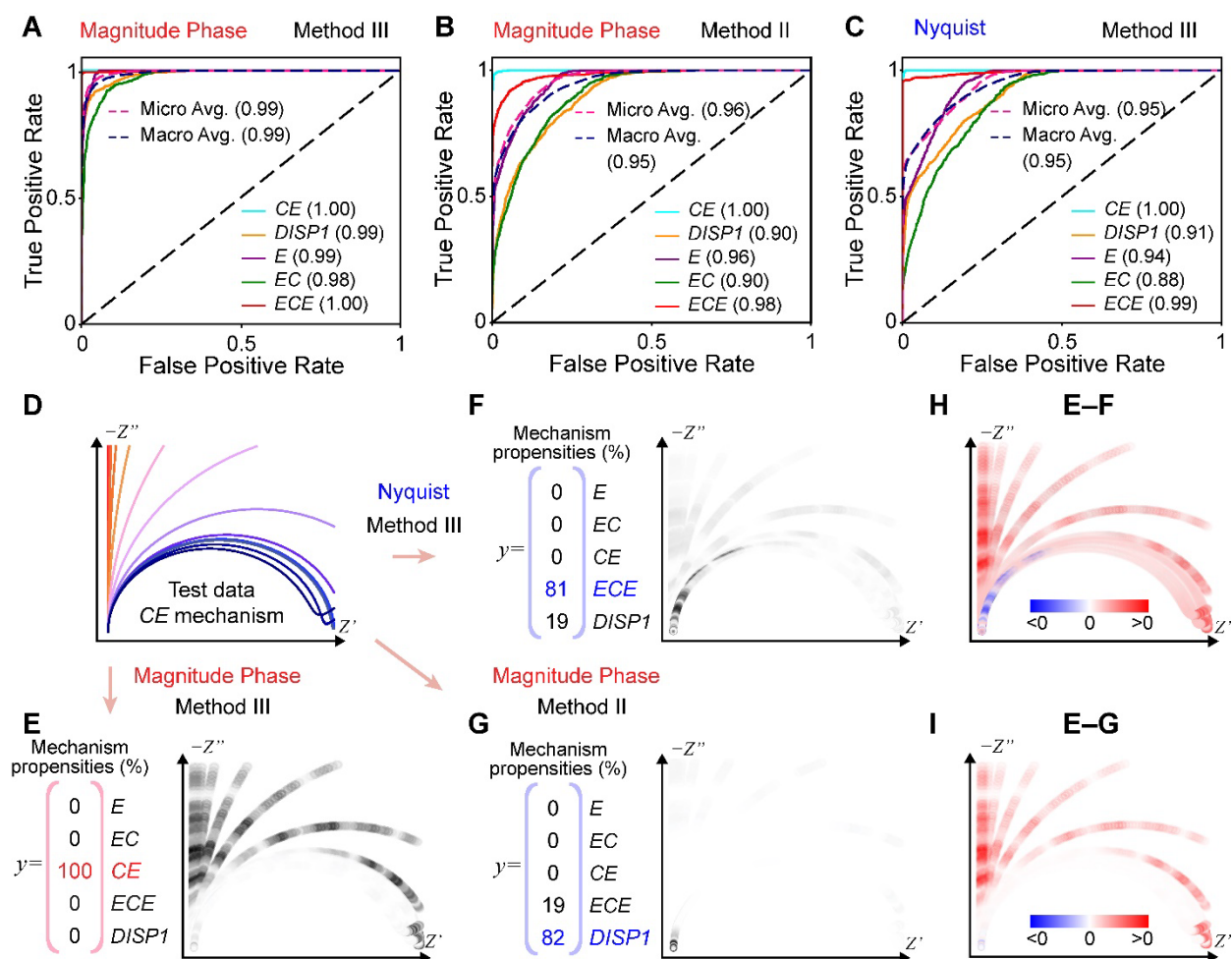


Figure 3. Comparative performance and “importance” evaluation of models with different preprocessing strategies. (A) to (C) Receiver operating characteristic (ROC) curves of the optimal model—magnitude-vs-phase representation and method III normalization, and two suboptimal models—magnitude-vs-phase representation with method II normalization and Nyquist representation with method III normalization. (D) Representative test data of CE mechanism, color-coded for different bias potentials. (E, F, and G) Predictions and “importance” evaluation results from models with varied preprocessing steps. (H) “Importance” discrepancy between models trained with magnitude-vs-phase and Nyquist representation, both using method III normalization. (I) “Importance” discrepancy between models with method III and method II normalization, both using magnitude-vs-phase representation. Red sections indicate higher “importance” in the optimal model, while blue regions signify higher “importance” in the other model.

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